

Rename:

```
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ sudo hostnamectl set-hostname ibrahim365
[sudo] password for ibrahim365:
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ sudo nano /etc/hosts
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ hostnamectl
Command 'hostnamectl' not found, did you mean:
  command 'hostnamectl' from deb systemd (255.4-1ubuntu8.16)
Try: sudo apt install <deb name>
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ hostnamectl
Command 'hostnamectl' not found, did you mean:
  command 'hostnamectl' from deb systemd (255.4-1ubuntu8.16)
Try: sudo apt install <deb name>
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ hostname
ibrahim365
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$
```

Upgrade System:

```

brahms08@brahms08:~$ sudo apt update
Hit:1 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
69 packages can be upgraded. Run 'apt list --upgradable' to see them.
brahms08@brahms08:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done

The following package was automatically installed and is no longer required:
libfswap2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
libfswap3
The following packages will be upgraded:
elfutils elfutils-common cloud-init virtutils fwupd gdb gdb3 glib2.0 adn-1.0 glib2.0 gtk-4.0 glib2.0 mutter-14 glib2.0 nm-1.0
gnome-control-center-gio gnome-control-center-faces gnome-keyring gnome-keyring-pkcs11 gnome-shell gnome-shell-common lib
libprinter-2.2 libprinter-2.0 radi libwpd3 libwpd4 libwpd5 libwpd6 libwpd7 libwpd8 libwpd9 libwpd10 libwpd11 libwpd12 libwpd13 libwpd14 libwpd15 libwpd16 libwpd17 libwpd18 libwpd19 libwpd20 libwpd21 libwpd22 libwpd23 libwpd24 libwpd25 libwpd26 libwpd27 libwpd28 libwpd29 libwpd30 libwpd31 libwpd32 libwpd33 libwpd34 libwpd35 libwpd36 libwpd37 libwpd38 libwpd39 libwpd40 libwpd41 libwpd42 libwpd43 libwpd44 libwpd45 libwpd46 libwpd47 libwpd48 libwpd49 libwpd50 libwpd51 libwpd52 libwpd53 libwpd54 libwpd55 libwpd56 libwpd57 libwpd58 libwpd59 libwpd60 libwpd61 libwpd62 libwpd63 libwpd64 libwpd65 libwpd66 libwpd67 libwpd68 libwpd69 libwpd70 libwpd71 libwpd72 libwpd73 libwpd74 libwpd75 libwpd76 libwpd77 libwpd78 libwpd79 libwpd80 libwpd81 libwpd82 libwpd83 libwpd84 libwpd85 libwpd86 libwpd87 libwpd88 libwpd89 libwpd90 libwpd91 libwpd92 libwpd93 libwpd94 libwpd95 libwpd96 libwpd97 libwpd98 libwpd99 libwpd100 libwpd101 libwpd102 libwpd103 libwpd104 libwpd105 libwpd106 libwpd107 libwpd108 libwpd109 libwpd110 libwpd111 libwpd112 libwpd113 libwpd114 libwpd115 libwpd116 libwpd117 libwpd118 libwpd119 libwpd120 libwpd121 libwpd122 libwpd123 libwpd124 libwpd125 libwpd126 libwpd127 libwpd128 libwpd129 libwpd130 libwpd131 libwpd132 libwpd133 libwpd134 libwpd135 libwpd136 libwpd137 libwpd138 libwpd139 libwpd140 libwpd141 libwpd142 libwpd143 libwpd144 libwpd145 libwpd146 libwpd147 libwpd148 libwpd149 libwpd150 libwpd151 libwpd152 libwpd153 libwpd154 libwpd155 libwpd156 libwpd157 libwpd158 libwpd159 libwpd160 libwpd161 libwpd162 libwpd163 libwpd164 libwpd165 libwpd166 libwpd167 libwpd168 libwpd169 libwpd170 libwpd171 libwpd172 libwpd173 libwpd174 libwpd175 libwpd176 libwpd177 libwpd178 libwpd179 libwpd180 libwpd181 libwpd182 libwpd183 libwpd184 libwpd185 libwpd186 libwpd187 libwpd188 libwpd189 libwpd190 libwpd191 libwpd192 libwpd193 libwpd194 libwpd195 libwpd196 libwpd197 libwpd198 libwpd199 libwpd200 libwpd201 libwpd202 libwpd203 libwpd204 libwpd205 libwpd206 libwpd207 libwpd208 libwpd209 libwpd210 libwpd211 libwpd212 libwpd213 libwpd214 libwpd215 libwpd216 libwpd217 libwpd218 libwpd219 libwpd220 libwpd221 libwpd222 libwpd223 libwpd224 libwpd225 libwpd226 libwpd227 libwpd228 libwpd229 libwpd230 libwpd231 libwpd232 libwpd233 libwpd234 libwpd235 libwpd236 libwpd237 libwpd238 libwpd239 libwpd240 libwpd241 libwpd242 libwpd243 libwpd244 libwpd245 libwpd246 libwpd247 libwpd248 libwpd249 libwpd250 libwpd251 libwpd252 libwpd253 libwpd254 libwpd255 libwpd256 libwpd257 libwpd258 libwpd259 libwpd260 libwpd261 libwpd262 libwpd263 libwpd264 libwpd265 libwpd266 libwpd267 libwpd268 libwpd269 libwpd270 libwpd271 libwpd272 libwpd273 libwpd274 libwpd275 libwpd276 libwpd277 libwpd278 libwpd279 libwpd280 libwpd281 libwpd282 libwpd283 libwpd284 libwpd285 libwpd286 libwpd287 libwpd288 libwpd289 libwpd290 libwpd291 libwpd292 libwpd293 libwpd294 libwpd295 libwpd296 libwpd297 libwpd298 libwpd299 libwpd300 libwpd301 libwpd302 libwpd303 libwpd304 libwpd305 libwpd306 libwpd307 libwpd308 libwpd309 libwpd310 libwpd311 libwpd312 libwpd313 libwpd314 libwpd315 libwpd316 libwpd317 libwpd318 libwpd319 libwpd320 libwpd321 libwpd322 libwpd323 libwpd324 libwpd325 libwpd326 libwpd327 libwpd328 libwpd329 libwpd330 libwpd331 libwpd332 libwpd333 libwpd334 libwpd335 libwpd336 libwpd337 libwpd338 libwpd339 libwpd340 libwpd341 libwpd342 libwpd343 libwpd344 libwpd345 libwpd346 libwpd347 libwpd348 libwpd349 libwpd350 libwpd351 libwpd352 libwpd353 libwpd354 libwpd355 libwpd356 libwpd357 libwpd358 libwpd359 libwpd360 libwpd361 libwpd362 libwpd363 libwpd364 libwpd365 libwpd366 libwpd367 libwpd368 libwpd369 libwpd370 libwpd371 libwpd372 libwpd373 libwpd374 libwpd375 libwpd376 libwpd377 libwpd378 libwpd379 libwpd380 libwpd381 libwpd382 libwpd383 libwpd384 libwpd385 libwpd386 libwpd387 libwpd388 libwpd389 libwpd390 libwpd391 libwpd392 libwpd393 libwpd394 libwpd395 libwpd396 libwpd397 libwpd398 libwpd399 libwpd400 libwpd401 libwpd402 libwpd403 libwpd404 libwpd405 libwpd406 libwpd407 libwpd408 libwpd409 libwpd410 libwpd411 libwpd412 libwpd413 libwpd414 libwpd415 libwpd416 libwpd417 libwpd418 libwpd419 libwpd420 libwpd421 libwpd422 libwpd423 libwpd424 libwpd425 libwpd426 libwpd427 libwpd428 libwpd429 libwpd430 libwpd431 libwpd432 libwpd433 libwpd434 libwpd435 libwpd436 libwpd437 libwpd438 libwpd439 libwpd440 libwpd441 libwpd442 libwpd443 libwpd444 libwpd445 libwpd446 libwpd447 libwpd448 libwpd449 libwpd450 libwpd451 libwpd452 libwpd453 libwpd454 libwpd455 libwpd456 libwpd457 libwpd458 libwpd459 libwpd460 libwpd461 libwpd462 libwpd463 libwpd464 libwpd465 libwpd466 libwpd467 libwpd468 libwpd469 libwpd470 libwpd471 libwpd472 libwpd473 libwpd474 libwpd475 libwpd476 libwpd477 libwpd478 libwpd479 libwpd480 libwpd481 libwpd482 libwpd483 libwpd484 libwpd485 libwpd486 libwpd487 libwpd488 libwpd489 libwpd490 libwpd491 libwpd492 libwpd493 libwpd494 libwpd495 libwpd496 libwpd497 libwpd498 libwpd499 libwpd500 libwpd501 libwpd502 libwpd503 libwpd504 libwpd505 libwpd506 libwpd507 libwpd508 libwpd509 libwpd510 libwpd511 libwpd512 libwpd513 libwpd514 libwpd515 libwpd516 libwpd517 libwpd518 libwpd519 libwpd520 libwpd521 libwpd522 libwpd523 libwpd524 libwpd525 libwpd526 libwpd527 libwpd528 libwpd529 libwpd530 libwpd531 libwpd532 libwpd533 libwpd534 libwpd535 libwpd536 libwpd537 libwpd538 libwpd539 libwpd540 libwpd541 libwpd542 libwpd543 libwpd544 libwpd545 libwpd546 libwpd547 libwpd548 libwpd549 libwpd550 libwpd551 libwpd552 libwpd553 libwpd554 libwpd555 libwpd556 libwpd557 libwpd558 libwpd559 libwpd560 libwpd561 libwpd562 libwpd563 libwpd564 libwpd565 libwpd566 libwpd567 libwpd568 libwpd569 libwpd570 libwpd571 libwpd572 libwpd573 libwpd574 libwpd575 libwpd576 libwpd577 libwpd578 libwpd579 libwpd580 libwpd581 libwpd582 libwpd583 libwpd584 libwpd585 libwpd586 libwpd587 libwpd588 libwpd589 libwpd590 libwpd591 libwpd592 libwpd593 libwpd594 libwpd595 libwpd596 libwpd597 libwpd598 libwpd599 libwpd600 libwpd601 libwpd602 libwpd603 libwpd604 libwpd605 libwpd606 libwpd607 libwpd608 libwpd609 libwpd610 libwpd611 libwpd612 libwpd613 libwpd614 libwpd615 libwpd616 libwpd617 libwpd618 libwpd619 libwpd620 libwpd621 libwpd622 libwpd623 libwpd624 libwpd625 libwpd626 libwpd627 libwpd628 libwpd629 libwpd630 libwpd631 libwpd632 libwpd633 libwpd634 libwpd635 libwpd636 libwpd637 libwpd638 libwpd639 libwpd640 libwpd641 libwpd642 libwpd643 libwpd644 libwpd645 libwpd646 libwpd647 libwpd648 libwpd64
```

Install JAVA:

```

root@kali:~/kali-Virtual-Box# ./kali-apt-get-install-openssh-ll-gpg-eg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

The following package was automatically installed and is no longer required:
  libfuzzy2
Use 'dpkg --get-selections' to remove it.

The following additional packages will be installed:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-gtk libice-dev libpthread-stubs-dev libxml-dev libxml2-dev
  libxmu-dev libxmu6 dev libxmu-dev libxmu6 openssh-client openssh-client-gtk openssh-client-gtk-headless x11proto-dev xorg-apt-setup xorg-docs_etc xtrans-dev
Suggested packages:
  default-jre libice0 libxml2 libxmu6 libxmu-dev libxmu6 openssh-client-gtk openssh-client-gtk-headless x11proto-dev xorg-apt-setup xorg-docs_etc xtrans-dev
  fonts-dejavu-core fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra
  fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra fonts-dejavu-extra
The following NEW packages will be installed:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-gtk libice-dev libpthread-stubs-dev libxml-dev libxml2-dev
  libxmu-dev libxmu6 dev libxmu-dev libxmu6 openssh-client openssh-client-gtk openssh-client-gtk-headless x11proto-dev xorg-apt-setup xorg-docs_etc xtrans-dev
0 upgraded, 20 newly installed, 0 to remove and 0 not upgraded.
Need to get 164 MB of archives.
After this operation, 219 MB of additional disk space will be used.
Get:1 http://ftp.kali.debian.org/debian/pool/main/o/openssh-client-gtk/openssh-client-gtk_7.9p1-4+b1_i386.deb [11.6 kB]

```

Install OpenSSH:

```
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  nelly-guard monkeysphere ssh-addpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 832 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu1.16 [37.3 kB]
Get:2 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu1.16 [510 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+20240113-ubuntu2 [275 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 ssh-import-id all 5.11-ubuntu2.24.04.1 [10.1 kB]
Fetched 832 kB in 2s (288 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 153650 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1:9.6p1-3ubuntu1.16_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu1.16) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu1.16_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu1.16) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-ubuntu2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-ubuntu2.24.04.1_all.deb ...
Unpacking ssh-import-id (5.11-ubuntu2.24.04.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu1.16) ...
```

```
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-21 14:03:46 PKT; 32s ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 12645 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 12646 (sshd)
     Tasks: 1 (limit: 4541)
    Memory: 1.2M (peak: 1.5M)
       CPU: 18ms
    CGroup: /system.slice/ssh.service
            └─12646 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 21 14:03:46 ibrahim365 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 21 14:03:46 ibrahim365 sshd[12646]: Server listening on 0.0.0.0 port 22.
Jun 21 14:03:46 ibrahim365 sshd[12646]: Server listening on :: port 22.
Jun 21 14:03:46 ibrahim365 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Create a group for the Hadoop user (Organizing user permissions):

```
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ sudo addgroup hadoop
info: Selecting GID from range 1000 to 59999 ...
info: Adding group 'hadoop' (GID 1001) ...
```

Create a Hadoop user and add it to the created group (A separate user for Hadoop security):

```
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ sudo adduser --ingroup hadoop hadoopuser
info: Adding user 'hadoopuser' ...
info: Selecting UID from range 1000 to 59999 ...

info: Adding new user 'hadoopuser' (1001) with group 'hadoop (1001)' ...
info: Creating home directory '/home/hadoopuser' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for hadoopuser
Enter the new value, or press ENTER for the default
    Full Name []: hadoopuser
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user 'hadoopuser' to supplemental / extra groups 'users' ...
info: Adding user 'hadoopuser' to group 'users' ...
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ sudo adduser hadoop sudo
fatal: The user 'hadoop' does not exist.
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ sudo adduser hadoopuser sudo
info: Adding user 'hadoopuser' to group 'sudo' ...
```

Assign permissions (Grant necessary permissions for Hadoop directories):

```
ibrahim365@ibrahim365-VMware-Virtual-Platform:~$ su - hadoopuser
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hadoopuser@ibrahim365:~$
```

Generate SSH keys and configure autologin (Essential for running Hadoop in pseudo-distributed mode):

```
hadoopuser@ibrahim365:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoopuser/.ssh/id_rsa):
Created directory '/home/hadoopuser/.ssh'.
Your identification has been saved in /home/hadoopuser/.ssh/id_rsa
Your public key has been saved in /home/hadoopuser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:VKERklqPU3DlABzS2X4M3GuMYlPDcnGptQnUNFp4bV4 hadoopuser@ibrahim365
The key's randomart image is:
+----[RSA 3072]-----+
|      ..==0=+o      |
|      +o@oO*E      |
|      .X @ooo=      |
|      + X.* o.*      |
|      o =So o      |
|                    |
+----+-----+
+-----[SHA256]-----+
```

```
hadoopuser@ibrahim365:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
hadoopuser@ibrahim365:~$ chmod 600 ~/.ssh/authorized_keys
hadoopuser@ibrahim365:~$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:nF8keUPQFXLlPKMz7R32xnDy0l3o67FBH5e8jAqScSA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.8-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

4 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hadoopuser@ibrahim365:~$ exit
Logout
Connection to localhost closed.
```


Download and copy Hadoop to the installation directory (Prepare Hadoop binaries):

```
hadoopuser@ibrahim365:~$ ls
hadoop-3.3.6  hadoop-3.3.6.tar.gz  snap
hadoopuser@ibrahim365:~$ sudo mv hadoop-3.3.6 /usr/local/hadoop
[sudo] password for hadoopuser:
hadoopuser@ibrahim365:~$ sudo chown -R hadoopuser:hadoop /usr/local/hadoop
hadoopuser@ibrahim365:~$ ls /usr/local
bin  etc  games  hadoop  include  lib  man  sbin  share  src
hadoopuser@ibrahim365:~$
```

```
hadoopuser@ibrahim365:~$ ls
hadoop-3.3.6  hadoop-3.3.6.tar.gz  snap
hadoopuser@ibrahim365:~$ sudo mv hadoop-3.3.6 /usr/local/hadoop
[sudo] password for hadoopuser:
hadoopuser@ibrahim365:~$ sudo chown -R hadoopuser:hadoop /usr/local/hadoop
hadoopuser@ibrahim365:~$ ls /usr/local
bin  etc  games  hadoop  include  lib  man  sbin  share  src
hadoopuser@ibrahim365:~$ nano ~/.bashrc
hadoopuser@ibrahim365:~$ source ~/.bashrc
hadoopuser@ibrahim365:~$ echo $JAVA_HOME
/usr/lib/jvm/java-11-openjdk-amd64
hadoopuser@ibrahim365:~$ echo $HADOOP_HOME
/usr/local/hadoop
hadoopuser@ibrahim365:~$
```

Configure core-site.xml (Set up Hadoop's basic configurations):

```
hadoopuser@ibrahim365:~$ nano /usr/local/hadoop/etc/hadoop/core-site.xml
```

```
GNU nano 7.2 /usr/local/hadoop/etc/had
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

Configure hdfs-site.xml (Set up HDFS properties):

```
hadoopuser@ibrahim365:~$ nano /usr/local/hadoop/etc/hadoop/hdfs-site.xml
```

```

GNU nano 7.2 /usr/local/hadoop/etc/hadoop/hd
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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you may not use this file except in compliance with the License.
You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/hadoopuser/hadoopdata/hdfs/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///home/hadoopuser/hadoopdata/hdfs/datanode</value>
  </property>

```

Configure mapred-site.xml (Set up MapReduce framework):

```
hadoopuser@ibrahim365:~$ nano /usr/local/hadoop/etc/hadoop/mapred-site.xml
```

```
GNU nano 7.2 /usr/local/hadoop/etc/hadoop/mapred-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

  <property>
    <name>yarn.app.mapreduce.am.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>

  <property>
    <name>mapreduce.map.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
</configuration>
```


Configure yarn-site.xml (Set up YARN resource manager):

```
hadoopuser@ibrahim365:~$ nano /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

```
GNU nano 7.2 /usr/local/hadoop/etc/hadoop/yarn-site.xml
<?xml version="1.0"?>
<!--
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you may not use this file except in compliance with the License.
You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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-->
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>

  <property>
    <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
    <value>org.apache.hadoop.mapred.ShuffleHandler</value>
  </property>
</configuration>
```

Format the HDFS Namenode (Mandatory step before starting services):

[illegible]

Start Hadoop services (start-dfs.sh and start-yarn.sh):

```
hadoopuser@ibrahim365:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [localhost]
ibrahim365:~$ jps
16100 NameNode
16276 DataNode
16725 ResourceManager
17238 Jps
16519 SecondaryNameNode
16846 NodeManager
hadoopuser@ibrahim365:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanager
hadoopuser@ibrahim365:~$
```

Verify running Hadoop services (jps command to check daemons):

```
hadoopuser@ibrahim365:~$ jps
16100 NameNode
16276 DataNode
16725 ResourceManager
17238 Jps
16519 SecondaryNameNode
16846 NodeManager
hadoopuser@ibrahim365:~$
```

Overview "localhost:9000" (active)

STARTED:	Sun Jun 21 14:30:27 +0303 2020
VERSION:	3.13.0, 1.3.0-202006180011, 3.13.0-202006180011, 3.13.0
COMPLETED:	Sun Jun 18 13:25:09 +0303 2020 by user@ibrahim365: [HADO] detached at release 3.13.0-RC1
Cluster ID:	010-153ba7fa-988a-4d13-a917-31a91a1a1a00
Block Pool ID:	BP-429033713-127.0.0.1-1702035749072

Summary

Security is off
Telemetry is off

1 files and directories, 0 blocks (replicated blocks), 0 erasure coded block groups = 1 total filesystem objects

Heap Memory used: 24.9 MB of 159 MB Heap Memory. Max heap Memory is 968 MB.

Non-Heap Memory used: 31.11 MB of 54.62 MB Committed Non-Heap Memory. Max Non-Heap Memory is Unbounded.

Configured Capacity:	24 GB
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The screenshot shows the Hadoop web interface at <http://localhost:8087/cluster>. The interface includes a sidebar with navigation links for Cluster, Tools, and a list of metrics (NEW, NEW, GIVING, GIVING, ACCEPTED, ACCEPTED, RUNNING, RUNNING, FAILED, FAILED, KILLED). The main content area displays Cluster Metrics, Cluster Nodes Metrics, Scheduler Metrics, and a table of running applications.

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State
Showing 0 to 0 of 0 entries										

Download a sample WordCount file for testing (Prepare input for WordCount example):



```
hadoopuser@ibrahim365:~$ sudo ls /home/ibrahim365/Downloads
hadoop.txt  wordcount.jar
```

Create an assignment folder and place the input file (Organize input data):

```
hadoopuser@ibrahim365:~$ mkdir -p /Assignment
hadoopuser@ibrahim365:~$ cp -r /hadoop.txt /Assignment/
hadoopuser@ibrahim365:~$ ls -l /Assignment
-rw-r--r-- 1 hadoopuser hadoopuser 1024 2023-10-27 10:27 hadoop.txt
hadoopuser@ibrahim365:~$
```

Run WordCount jar example on input file (hadoop jar command)

```
2026-06-21 17:49:15,735 INFO mapreduce.Job: map 0% reduce 0%
2026-06-21 17:49:31,936 INFO mapreduce.Job: map 52% reduce 0%
2026-06-21 17:49:38,638 INFO mapreduce.Job: map 81% reduce 0%
2026-06-21 17:49:39,069 INFO mapreduce.Job: map 100% reduce 0%
2026-06-21 17:49:45,117 INFO mapreduce.Job: map 100% reduce 100%
2026-06-21 17:49:47,138 INFO mapreduce.Job: Job job_1782015830351_0002 completed successfully
2026-06-21 17:49:47,245 INFO mapreduce.Job: Counters: 54
```

View the output (hdfs dfs -cat output_path/part-r-00000):

```
55555 11
Directed      1
Edges: 1
FromNodeId    1
Google 1
Nodes: 1
ToNodeId      1
Webgraph      1
contest,      1
from 1
graph 1
is 1
nodes 1
of 1
once): 1
pair 1
programming 1
saved 1
the 1
unordered 1
web-Google.txt 1
hadoopuser@ibrahim365:~$
```